Paraphraser Fingerprinting

Can We Identify the Rewriter from the Text Alone?

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Research Question

RQ: Paraphraser Fingerprints

Do different paraphrasing models leave distinct, detectable traces?
Can we identify the paraphraser from the text alone?

- 4 paraphrasers: Llama-3.1-8B, Gemini-2.5-Flash, GPT-5-Nano, DIPPER
- Detective (Claude Sonnet 4.6) matches outputs to paraphraser
- Auditor (Claude Sonnet 4.6) iteratively crafts adversarial source texts

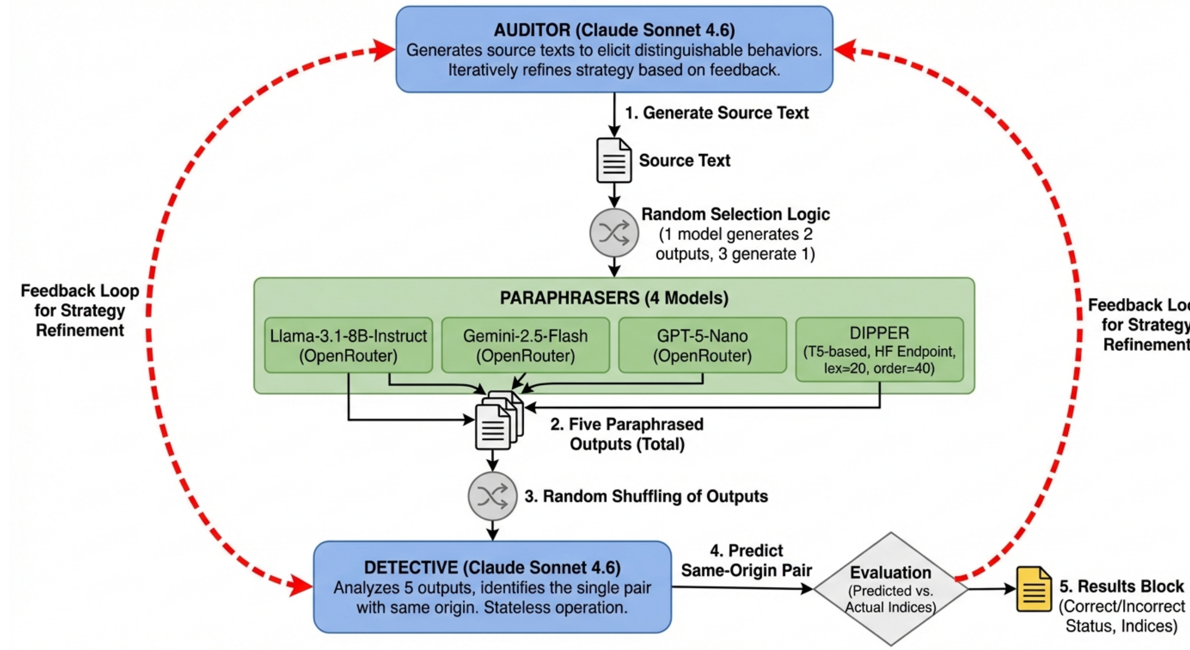
Dataset & Trial Protocol

20 trials per run

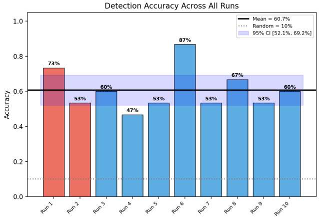
- Trial 1 only: hand-crafted text with linguistic traps (grammar, repetition, temporal disorder)
- Trials 2–20: Auditor-generated via APE-style iterative refinement

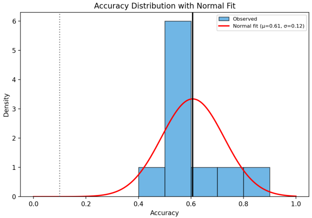
Each trial:

1. Auditor generates source text
2. 1 random paraphraser produces 2 outputs; other 3 produce 1 each → 5 total
3. Outputs shuffled
4. Detective identifies the matching pair
5. Result fed back to Auditor


AUDITOR (Claude Sonnet 4.6)
Generates source texts to elicit distinguishable behaviors. Iteratively refines strategy based on feedback.
1. Generate Source Text
Source Text
Random Selection Logic
(1 model generates 2 outputs, 3 generate 1)
Feedback Loop for Strategy Refinement
PARAPHRASERS (4 Models)
Llama-3.1-8B-Instruct (OpenRouter)
Gemini-2.5-Flash (OpenRouter)
GPT-5-Nano (OpenRouter)
DIPPER (T5-based, HF Endpoint, lex=20, order=40)
2. Five Paraphrased Outputs (Total)
3. Random Shuffling of Outputs
DETECTIVE (Claude Sonnet 4.6)
Analyzes 5 outputs, identifies the single pair with same origin. Stateless operation.
4. Predict Same-Origin Pair
Evaluation
(Predicted vs. Actual Indices)
5. Results Block
(Correct/Incorrect Status, Indices)

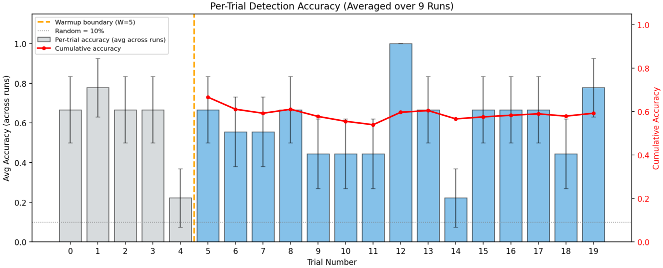
Aggregate Accuracy





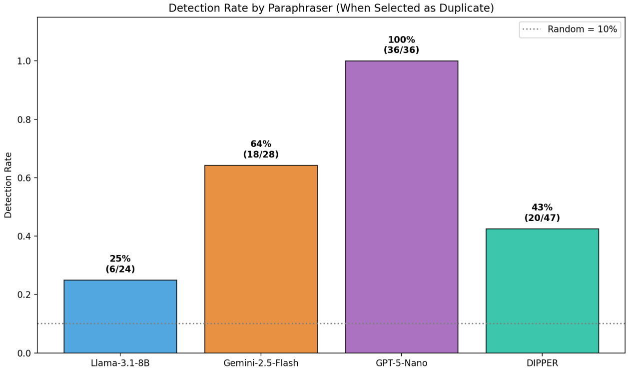
- **Mean: 60.7%** across 10 runs (range 47–87%)
- Random baseline: 10%
- Statistically significant: $t(9) = 13.41$, $p < 0.001$
- Detective reliably above chance

Per-Trial Learning Curve



- Trials 1–5: warm-up
- Red curve = cumulative accuracy from trial 5+
- Auditor's APE refinement visible in later trials

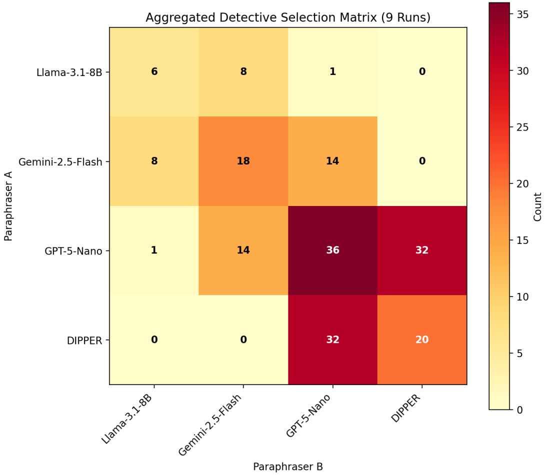
Per-Paraphraser Detection Rates



- **GPT-5-Nano**: **100%** (36/36)
- **Gemini-2.5-Flash**: 64.3% (18/28)
- **DIPPER**: 42.6% (20/47)
- **Llama-3.1-8B**: 25.0% (6/24)

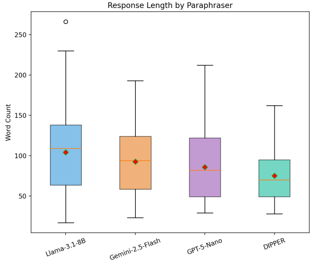
Fingerprint strength varies dramatically across models.

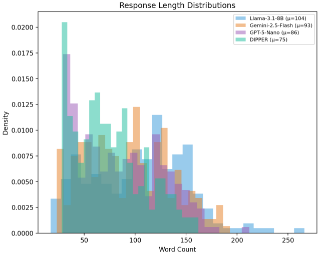
Confusion Analysis



- **GPT-5-Nano** ↔ **DIPPER** (32): most confused pair despite different architectures
- Gemini ↔ GPT-5-Nano (14): moderate LLM-LLM confusion
- **DIPPER** ↔ **Llama**: never confused (0)

Response Length as Fingerprint

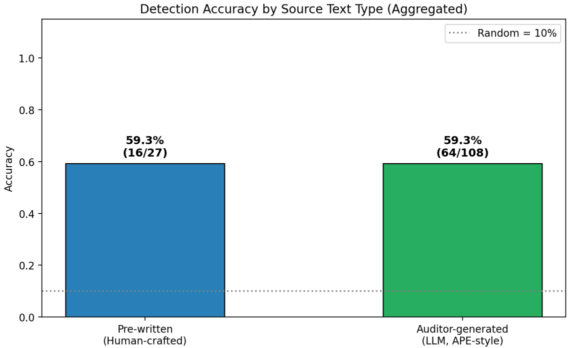




Model	**Mean (w)**
Llama-3.1-8B	104
Gemini-2.5-Flash	93
GPT-5-Nano	86
DIPPER	75

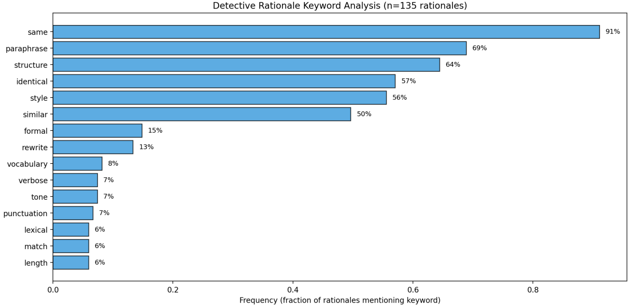
- GPT-5-Nano: moderate length but **strongest fingerprint**
- Length alone insufficient—style matters more

Source Text Type Comparison



- Human-crafted: **59.3%**
- Auditor-generated: **59.3%**
- APE-style refinement matches human-designed traps
- Validates automated source generation strategy

Detective Rationale Analysis



Top features ($n = 135$ rationales):

- **Structure**: 64%
- **Style**: 56%
- Formal: 15%
- Vocabulary: 8%
- Tone: 7%
- Length: 6%

Structural/stylistic features dominate over surface heuristics.

Next Steps

1. **Stronger models** — Test Claude Opus / GPT-5 as Detective/Auditor
2. **Expand paraphraser pool** — Mistral, Command-R, Qwen; same-family comparisons (Llama-8B vs. 70B)

Conclusion

- Mean detection: **60.7%** vs. 10% random baseline ($p < 0.001$)
- Fingerprint strength varies widely: GPT-5-Nano (100%) $\rightarrow$ Llama (25%)
- Detective relies on **structure & style**, not just output length
- APE-generated source texts match hand-crafted trap effectiveness

Takeaway

The "wood grain" of AI models is real and measurable—paraphraser attribution is feasible.